

The Mechanism of the Acoustic Radiation Force on a Free Elastic Cylinder in a Bounded Viscous Fluid

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ABSTRACT

This paper deals with analytical and numerical investigations of the acoustic radiation force (ARF) on a free cylinder in a bounded viscous fluid under oblique plane wave incidence. The scattering coefficients of a free elastic cylinder in that sound field defined in the paper are calculated theoretically. The fluid viscosity, plane wave's incident angle, and particle position are considered in the ARF problems. Results show that the mechanism of the ARF for a free elastic cylinder is dependent on the various factors considered in this paper. Compared with the ARFs for the particle in the high viscosity liquid (glycerin), the ARFs versus kR for the particle in the low viscosity liquid (water) appear a series of sharp peaks. This work contributes to better understanding of the acoustic manipulation mechanism using ARF.

Keywords: acoustic radiation force; viscous fluid; free elastic cylinder; boundary

1. INTRODUCTION

The acoustic radiation force (ARF), as a well-defined scientific subject, began with the work of Rayleigh [1]. Shortly after, the ARF on bubbles investigated by Bjerknes [2]. The ARFs of particles were explored later in works carried out by King and Yosioka and Kawasima [3-4]. Based on their works, Significant developments have occurred in understanding and testing ARF [5-14]. Nowadays, the noncontact manipulation of small particles is becoming increasingly important. Compared with other forces suitable for particle manipulation, the ARF has a number of advantages, such as high biocompatibility, noninvasive, label-free operation, a wider range of particle types, spanning a larger scale of particle in length and so on. These advantages make acoustic manipulation using ARF applicable in diverse areas, including biophysics, ultrasonic medicine etc. [15-16]. To improve the accuracy and effectiveness of particle manipulation technology, the ARF of particles immersed in viscous fluids was studied, but with the assumption of axisymmetric sound field, fixed particle, low viscosity fluid, or unbounded region [17-21].

To find real applications, asymmetric sound field, arbitrary viscosity fluid, bounded sound field and free particles must be factored in. A rigid cylinder was assumed in our previous work to clearly illustrate viscosity effects [22]. In many practical applications, the material of particles is usually non-rigid. In this paper, the elastic particle suspended freely in a bounded viscous fluid is considered to make it closer to the actual application environment. Additionally, the scattering coefficients of an elastic cylinder are calculated theoretically. Furthermore, the effects of fluid viscosity, plane wave's incident angle, and particle position on the ARF of the free elastic cylinder are discussed and analyzed in numerical simulations.

2. THEORY

2.1 Expansions of Sound Waves

As schematically depicted in Fig. 1, a plane wave propagates in the θ direction. The range of θ is $-\pi/2$ to $\pi/2$. The velocity potential is formulated in the cylindrical coordinate system (r_1, ϕ_1) whose origin is located at the free cylinder as

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$$\Phi_{\text{int}} = A \sum_{n=-\infty}^{\infty} i^n J_n(\alpha r_1) e^{in(\varphi_1 - \theta)} e^{-i\alpha d \cos \theta} e^{-i\omega t} \quad (1)$$

Here, $J_n(\cdot)$ means the Bessel function of the first kind, ω and t represent the angular frequency and time, respectively. The amplitude is defined as

$$A = \sqrt{2I_0 / \rho_0 c_0 k^2}$$

where I_0 represents the incident wave's intensity, while ρ_0 , c_0 stand for the density and sound velocity of liquids, respectively. Here, $k = \text{Re}(\alpha)$. The longitudinal wave's wavenumber is given by

$$\alpha = \frac{\omega}{c_0} \left[1 - i\omega(\lambda' + 2\mu') / \rho_0 c_0^2 \right]^{-1/2}$$

where parameters $\lambda' = \eta' - 2\mu'/3$, and η' , μ' represent the second and dynamic viscosity coefficients separately.

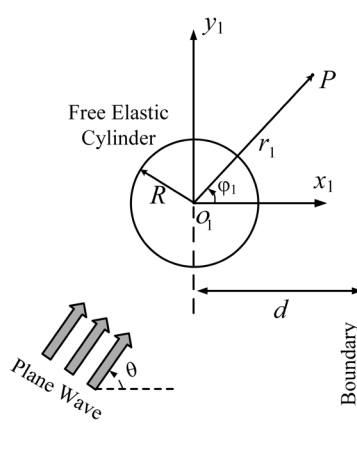


Figure 1. Schematic illustration of a free cylinder in a bounded viscous fluid under oblique plane wave incidence.

As a plane wave strikes the cylindrical particle, the scattering waves are generated and expressed in the system (r_1, φ_1) as [22]

$$\Phi_l^{(1)} = \sum_{n=-\infty}^{\infty} A_n^{(1)} H_n^{(1)}(\alpha r_1) e^{in\varphi_1} e^{-i\omega t} \quad (2)$$

$$\Psi_t^{(1)} = \sum_{n=-\infty}^{\infty} B_n^{(1)} H_n^{(1)}(\beta r_1) e^{in\varphi_1} e^{-i\omega t}, \quad (3)$$

where $\Phi_l^{(1)}$ denotes the potential scattered by the free cylinder for longitudinal wave, while $\Psi_t^{(1)}$ denotes that for transverse wave. $A_n^{(1)}$ and $B_n^{(1)}$, as scattering coefficients, are specified by conditions at the cylinder's lateral boundary. $H_n^{(1)}(\cdot) = J_n(\cdot) + iY_n(\cdot)$ means the Hankel function of the first kind. The shear wave's wavenumber is indicated by $\beta = (1+i)/\delta$. Here, the viscous wave's attenuation depth $\delta = \sqrt{2\mu' / \rho_0 \omega}$.

Sound waves continue to travel forward and are reflected at the flat boundary perpendicular to the wave's direction. This reflection is characterized using the method of images. An image cylinder and a mirrored plane wave are introduced with respect to the boundary. The mirror particle and the original particle have the same radius R and their distance from the boundary is d . Another cylindrical coordinate system (r_2, φ_2) is introduced and centered at the image particle. The

reflection problem of the boundary is transformed into the mirror incident wave and the scattering of the mirror particle. The potentials of the mirror incident wave and scattered waves are written in their respective coordinate systems:

$$\Phi_r = A \sum_{n=-\infty}^{\infty} i^n J_n(\alpha r_1) e^{-in(\pi-\varphi_1-\theta)} e^{i\alpha d \cos \theta} e^{-i\omega t} \quad (4)$$

$$\Phi_l^{(2)} = \sum_{n=-\infty}^{\infty} A_n^{(2)} H_n^{(1)}(\alpha r_2) e^{in\varphi_2} e^{-i\omega t} \quad (5)$$

$$\Psi_t^{(2)} = \sum_{n=-\infty}^{\infty} B_n^{(2)} H_n^{(1)}(\beta r_2) e^{in\varphi_2} e^{-i\omega t} \quad (6)$$

where Φ_r is the mirror incident wave, $\Phi_l^{(2)}$ and $\Psi_t^{(2)}$ represent longitudinal and transverse waves scattered by the image cylinder, $A_n^{(2)} = (-1)^n A_n^{(1)}$, and $B_n^{(2)} = (-1)^n B_n^{(1)}$.

Applying the addition theorem of cylindrical wave functions [23], (5) and (6) are expressed in the coordinate system (r_1, φ_1) as

$$\Phi_l^{(2)'} = \sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} (-1)^m A_m^{(1)} H_{n-m}^{(1)}(2\alpha d) J_n(\alpha r_1) e^{in\varphi_1} e^{-i\omega t} \quad (7)$$

$$\Psi_t^{(2)'} = \sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} (-1)^m B_m^{(1)} H_{n-m}^{(1)}(2\beta d) J_n(\beta r_1) e^{in\varphi_1} e^{-i\omega t}. \quad (8)$$

From (1) – (4) and (7) – (8), the overall velocity potential is expressed as

$$\begin{aligned} \Phi &= \Phi_i + \Phi_r + \Phi_l^{(1)} + \Phi_l^{(2)'} \\ &= \sum_{n=-\infty}^{\infty} \left[2A \cos \left(\alpha d \cos \theta + n\theta - \frac{n\pi}{2} \right) J_n(\alpha r_1) \right. \\ &\quad \left. + A_n^{(1)} H_n^{(1)}(\alpha r_1) + S_n J_n(\alpha r_1) \right] e^{in\varphi_1} e^{-i\omega t} \end{aligned} \quad (9)$$

$$\begin{aligned} \Psi &= \Psi_l^{(1)} + \Psi_l^{(2)'} \\ &= \sum_{n=-\infty}^{\infty} \left[B_n^{(1)} H_n^{(1)}(\beta r_1) + Q_n J_n(\beta r_1) \right] e^{in\varphi_1} e^{-i\omega t}, \end{aligned} \quad (10)$$

where

$$S_n = \sum_{m=-\infty}^{\infty} (-1)^m A_m^{(1)} H_{n-m}^{(1)}(2\alpha d),$$

$$Q_n = \sum_{m=-\infty}^{\infty} (-1)^m B_m^{(1)} H_{n-m}^{(1)}(2\beta d).$$

2.2 Scattering Field of a Free Elastic Cylinder

To calculate the ARF, the total velocity potential in the fluid given by (9) – (10) should be obtained, so the unknown scattering coefficient $A_n^{(1)}$ and $B_n^{(1)}$ must be calculated first.

For an infinitely long, isotropic elastic cylinder, when the sound wave is incident on it, the longitudinal wave and transverse waves will be generated in the particle. And, they are represented in the coordinate system (r_1, φ_1) by

$$\bar{\Phi}_l = \sum_{n=-\infty}^{\infty} \bar{A}_n J_n(\bar{k}_l r_1) e^{in\varphi_1} e^{-i\omega t}, \quad (11)$$

$$\bar{\Psi}_t = \sum_{n=-\infty}^{\infty} \bar{B}_n J_n(\bar{k}_t r_1) e^{in\varphi_1} e^{-i\omega t}, \quad (12)$$

Here, the longitudinal and transverse wavenumbers $\bar{k}_l$, $\bar{k}_t$ are represented by

$$\bar{k}_l = \frac{\omega}{c_l}$$

$$\bar{k}_t = \frac{\omega}{c_t}$$

where the wave velocities c_l and c_t are expressed by

$$c_l = \sqrt{\frac{\lambda + 2\mu}{\rho}}$$

$$c_t = \sqrt{\frac{\mu}{\rho}}$$

Here, λ and μ denote Lamé coefficients.

To calculate $A_n^{(1)}$ and $B_n^{(1)}$, the following boundary conditions are applied:

$$\begin{aligned} v_r \Big|_{r=R} &= \bar{v}_r \Big|_{r=R}, \\ v_\varphi \Big|_{r=R} &= \bar{v}_\varphi \Big|_{r=R}, \\ \sigma_{rr} \Big|_{r=R} &= \bar{\sigma}_{rr} \Big|_{r=R}, \\ \sigma_{r\varphi} \Big|_{r=R} &= \bar{\sigma}_{r\varphi} \Big|_{r=R}, \end{aligned} \quad (13)$$

where

$$\begin{aligned} \bar{v}_r &= \frac{\partial \bar{\Phi}_l}{\partial r} + \frac{1}{r} \frac{\partial \bar{\Psi}_t}{\partial \varphi}, \\ \bar{v}_\varphi &= \frac{1}{r} \frac{\partial \bar{\Phi}_l}{\partial \varphi} - \frac{\partial \bar{\Psi}_t}{\partial r}, \\ \bar{\sigma}_{rr} &= -\lambda k_l^2 \bar{\Phi}_l + 2\mu \frac{\partial \bar{v}_r}{\partial r}, \\ \bar{\sigma}_{r\varphi} &= \mu \left[\frac{1}{r} \frac{\partial \bar{v}_r}{\partial \varphi} + \frac{\partial \bar{v}_\varphi}{\partial r} - \frac{\bar{v}_\varphi}{r} \right], \end{aligned} \quad (14)$$

$$\begin{aligned}
v_r &= \frac{\partial \Phi}{\partial r} + \frac{1}{r} \frac{\partial \Psi}{\partial \varphi}, \\
v_\varphi &= \frac{1}{r} \frac{\partial \Phi}{\partial \varphi} - \frac{\partial \Psi}{\partial r}, \\
\sigma_{rr} &= -p_1 + 2\mu' \frac{\partial v_r}{\partial r} + \lambda' \left(\frac{\partial v_r}{\partial r} + \frac{1}{r} \frac{\partial v_\varphi}{\partial \varphi} + \frac{v_r}{r} \right), \\
\sigma_{r\varphi} &= \mu' \left(\frac{1}{r} \frac{\partial v_r}{\partial \varphi} + \frac{\partial v_\varphi}{\partial r} - \frac{v_\varphi}{r} \right),
\end{aligned} \tag{15}$$

in which

$$\begin{aligned}
p_1 &= \rho_0 \left(\frac{\lambda' + 2\mu'}{\rho_0} \Delta - \frac{\partial}{\partial t} \right) \Phi + \frac{\rho_0}{2c_0^2} \left(\frac{\partial \Phi}{\partial t} \right)^2 \\
&\quad - \frac{1}{2} \rho_0 \left(\nabla \Phi \right)^2 - \frac{\lambda' + 2\mu'}{c_0^2} \frac{\partial \Phi}{\partial t} \Delta \Phi.
\end{aligned} \tag{16}$$

Note that the partial derivative $\partial \Phi / \partial t$ in (16), in the case of a free cylinder, is evaluated using [22]

$$\frac{\partial \Phi}{\partial t} = \frac{d\Phi}{dt} - \mathbf{U} \cdot \nabla \Phi, \tag{17}$$

The velocity vector $\mathbf{U}$ is determined by

$$m \dot{\mathbf{U}} = \iint_{s_0} \boldsymbol{\sigma} d\mathbf{S}, \tag{18}$$

where the mass of a unit length cylinder is evaluated by $m = \pi R^2 \rho_1$, and ρ_1 represents the cylinder's density. The tensor of stresses is expressed as

$$\boldsymbol{\sigma} = (-p_1 + \lambda' \nabla \cdot \mathbf{v}) \mathbf{E} + 2\mu' \mathbf{e}$$

where $\mathbf{E}$ denotes a unit tensor, and the deformation velocity tensor is presented as

$$\mathbf{e} = \left((\nabla \mathbf{v}) + (\nabla \mathbf{v})^T \right) / 2$$

with the superscript T represents vector transposition.

Incorporating (9) – (12) into (13), after some arithmetic manipulations and sorting, a linear system of equations is obtained

$$\begin{bmatrix} C_r & D_r \\ C_\varphi & D_\varphi \end{bmatrix} - \begin{bmatrix} S_r & Q_r \\ S_\varphi & Q_\varphi \end{bmatrix} \begin{bmatrix} W_{1n} & O \\ O & W_{2n} \end{bmatrix} \begin{bmatrix} A_n \\ B_n \end{bmatrix} = \begin{bmatrix} I_r \\ I_\varphi \end{bmatrix}. \tag{19}$$

The explicit forms of the equations are provided in Appendix A. Solving Eq. (19) yields the unknown scattering coefficients.

2.3 ARF on a Free Cylinder in a Bounded Viscous Fluid

The ARF on a freely suspended cylinder in a viscous fluid is [17]

$$\mathbf{F} = \left\langle \iint_{s_0} \boldsymbol{\sigma} d\mathbf{S} \right\rangle, \tag{20}$$

where $\langle \rangle$ represents the time average. The total ARF is given in the form of the axial F_x and transverse F_y components, namely, $\mathbf{F} = F_x \mathbf{e}_x + F_y \mathbf{e}_y$, in which, $\mathbf{e}_x$ and $\mathbf{e}_y$ are the unit vectors. Applying (9)–(10) and (15)–(18) to (20) yields F_x and F_y as follows [22]

$$\begin{aligned}
 F_x = & \frac{\eta \rho_0 \pi}{R} \left[-(G_1 + N_1)G_2 - (L_1 - M_1)L_2 + 2M_{-1}M_{-2} \right. \\
 & \left. + N_{-1}G_{-2} - G_{-1}G_{-2} - L_{-1}L_{-2} - M_{-1}L_{-2} + M_{-2}L_{-1} - N_{-2}G_{-1} \right] \\
 & - \frac{\pi R \rho_0 \omega^2}{4c_0^2} \sum_{n=-\infty}^{\infty} (G_n G_{n+1} + L_n L_{n+1} + G_n G_{n-1} + L_n L_{n-1}) \\
 & + \frac{\rho_0 \pi}{4R} \sum_{n=-\infty}^{\infty} n \left[(n+1)(N_n N_{n+1} + M_n M_{n+1}) + (n-1)(N_n N_{n-1} + M_n M_{n-1}) \right] \\
 & + \frac{\rho_0 \pi}{4R} \sum_{n=-\infty}^{\infty} n(n+1)(L_n L_{n+1} + G_n G_{n+1}) + n(n-1)(L_n L_{n-1} + G_n G_{n-1}) \\
 & - \frac{\pi R \rho_0^2 (\lambda' + 2\mu') \omega^3}{2 \left[\rho_0^2 c_0^2 + \omega^2 (\lambda' + 2\mu')^2 \right]} \sum_{n=-\infty}^{\infty} (L_n G_{n+1} + L_n G_{n-1} - L_{n+1} G_n - L_{n-1} G_n) \\
 & + \frac{\pi R \rho_0 (\lambda' + 2\mu')^2 \omega^4}{2c_0^2 \left[\rho_0^2 c_0^2 + \omega^2 (\lambda' + 2\mu')^2 \right]} \sum_{n=-\infty}^{\infty} (G_n G_{n+1} + L_n L_{n+1} + G_n G_{n-1} + L_n L_{n-1}),
 \end{aligned} \tag{21}$$

$$\begin{aligned}
 F_y = & \frac{\eta \rho_0 \pi}{R} \left[(L_2 G_1 - L_2 N_1) - (G_2 L_1 + G_2 M_1) + (G_{-2} L_{-1} + G_{-2} M_{-1}) - (L_{-2} N_{-1} + L_{-2} G_{-1}) \right] \\
 & - \frac{\pi R \rho_0 \omega^2}{4c_0^2} \sum_{n=-\infty}^{\infty} (L_n G_{n+1} - G_n L_{n+1} - L_n G_{n-1} + G_n L_{n-1}) \\
 & + \frac{\rho_0 \pi}{4R} \sum_{n=-\infty}^{\infty} n \left[(n+1)(-M_n N_{n+1} + N_n M_{n+1}) + (n-1)(M_n N_{n-1} - N_n M_{n-1}) \right] \\
 & + \frac{\rho_0 \pi}{4R} \sum_{n=-\infty}^{\infty} n \left[(n+1)(L_n G_{n+1} - G_n L_{n+1}) + (n-1)(G_n L_{n-1} - L_n G_{n-1}) \right] \\
 & - \frac{\pi R \rho_0^2 (\lambda' + 2\mu') \omega^3}{2 \left[\rho_0^2 c_0^2 + \omega^2 (\lambda' + 2\mu')^2 \right]} \sum_{n=-\infty}^{\infty} (-L_n L_{n+1} + L_n L_{n-1} - G_n G_{n+1} + G_n G_{n-1}) \\
 & + \frac{\pi R \rho_0 (\lambda' + 2\mu')^2 \omega^4}{2c_0^2 \left[\rho_0^2 c_0^2 + \omega^2 (\lambda' + 2\mu')^2 \right]} \sum_{n=-\infty}^{\infty} (L_n G_{n+1} - G_n L_{n+1} - L_n G_{n-1} + G_n L_{n-1}).
 \end{aligned} \tag{22}$$

where

$$\begin{aligned}
 G_n &= \text{Re} \left[2A \cos \left(\alpha d \cos \theta + n\theta - \frac{n\pi}{2} \right) J_n(\alpha r_1) + A_n^{(1)} H_n^{(1)}(\alpha r_l) + S_n J_n(\alpha r_1) \right] \\
 L_n &= \text{Im} \left[2A \cos \left(\alpha d \cos \theta + n\theta - \frac{n\pi}{2} \right) J_n(\alpha r_1) + A_n^{(1)} H_n^{(1)}(\alpha r_l) + S_n J_n(\alpha r_1) \right] \\
 M_n &= \text{Re} \left[B_n^{(1)} H_n^{(1)}(\beta r_l) + Q_n J_n(\beta r_1) \right] \\
 N_n &= \text{Im} \left[B_n^{(1)} H_n^{(1)}(\beta r_l) + Q_n J_n(\beta r_1) \right].
 \end{aligned}$$

3. NUMERICAL RESULTS AND DISCUSSIONS

In the following, the nature and general behavior of the ARFs on the free elastic cylinder in that sound field defined in the paper are illustrated intuitively. A plane wave with intensity [27] $I_0 = 175.5 W/m^2$ is incident. The radius of the free cylinder is $R=0.01mm$. If not mentioned separately, the particle radius is 0.01mm. A high viscosity liquid (glycerin) and a low viscosity liquid (water) are selected as the surrounding liquid and their parameters are given in Table I. The truncation constants of $N_{max}=M_{max}=30$ are chosen to ensure convergence. The physical parameters of the free cylinder are provided in Table II.

TABLE I. The parameters of viscous fluids [19].

<i>Material</i>	<i>Density ρ (kg/m³)</i>	<i>Sound speed c_0 (m/s)</i>	<i>Dynamic viscosity η (Pa·s)</i>
Water	1000	1500	0.001
Glycerin	1260	1900	1.48

TABLE II. The physical parameters of the free cylinder [9].

<i>Material</i>	<i>Density ρ (kg/m³)</i>	<i>Compressional velocity c_l (m/s)</i>	<i>Shear velocity c_t (m/s)</i>
Polyethylene (PE)	957	2430	950

3.1 Effects of the Incident Angle

The PE cylinder is taken as an example and the ARFs in a high viscosity liquid (glycerin) versus θ and d/R are shown in Fig. 2. The parameters $kR=0.1$ and $-\pi/2 \leq \theta \leq \pi/2$. When θ is zero, the incoming wave is normal to the boundary; for $|\theta| = \pi/2$, it is parallel. Figs. 2(a) and (b) represent the axial and transverse ARFs, respectively. The results display that F_x and F_y are symmetric about the axis $\theta = 0$, as they satisfy $F_x(\theta) = F_x(-\theta)$ and $F_y(\theta) = -F_y(-\theta)$, respectively. Meanwhile, at $\theta = 0$, the amplitude of F_x is maximum due to the strengthened reflection from the boundary; as a result of the sound field's symmetry, F_y becomes zero, while the amplitude reaches its maximum at $|\theta| = \pi/2$.

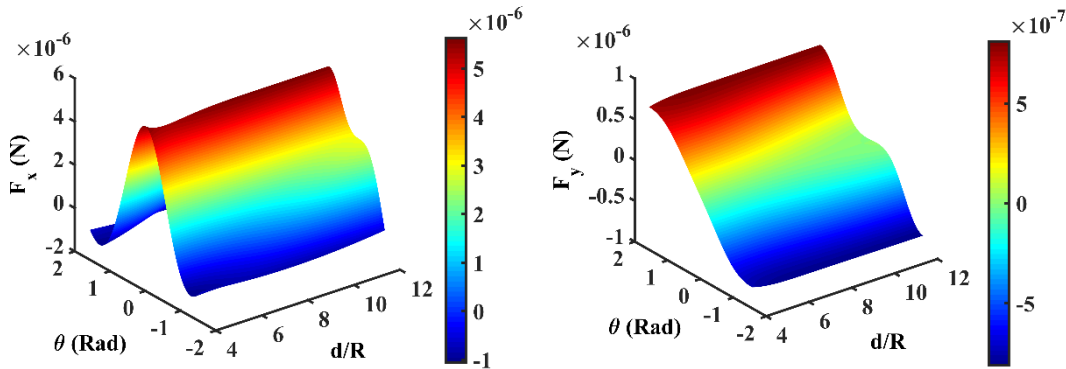


Figure 2. Variation of ARFs with θ and d/R for a free PE cylinder in glycerin at $kR=0.1$. (a) The axial ARF F_x , (b) The transverse ARF F_y .

3.2 ARFs of the Free Cylinder in Different Viscosity Liquid

Then, a high viscosity liquid (glycerin) and a low viscosity liquid (water) are selected as the surrounding liquid to analyze the effect of liquid viscosity on ARF. The ARFs on a PE particle freely suspended in the two liquids are calculated, respectively, and the results are exhibited in Fig. 3 and 4. Figs. 3(a) and 4(a) show the axial ARFs of particles

in glycerin and water, respectively. Figs. 3(b) and 4(b) present the transverse ARFs of particles in glycerin and water, respectively. The d are set as $2R$, $3R$, and $4R$ respectively. The $\theta = \pi/4$. The dimensionless frequency ranges from 0 to 5. From the figures, we find that the ARFs exhibit a series of maxima and minima because of the resonant vibrations of the cylinder, including its eigenmode oscillation and the motion. Compared with the ARFs for the particle in the high viscosity liquid (glycerin), the ARFs for the particle in the low viscosity liquid (water) appear a series of sharp peaks. In addition, the pronounced maxima and minima are mainly emerge on the larger kR . The reason is that the δ/R in glycerin or at a small kR is larger than that in water or at a large kR , a thick viscous boundary layer is formed on the particle surface. The thicker boundary layer leads to more dissipation and attenuation of acoustic waves, which leads to the broadening of the resonant peaks. In turn, the thin boundary layer does not cause noticeable broadening of the resonant peaks.

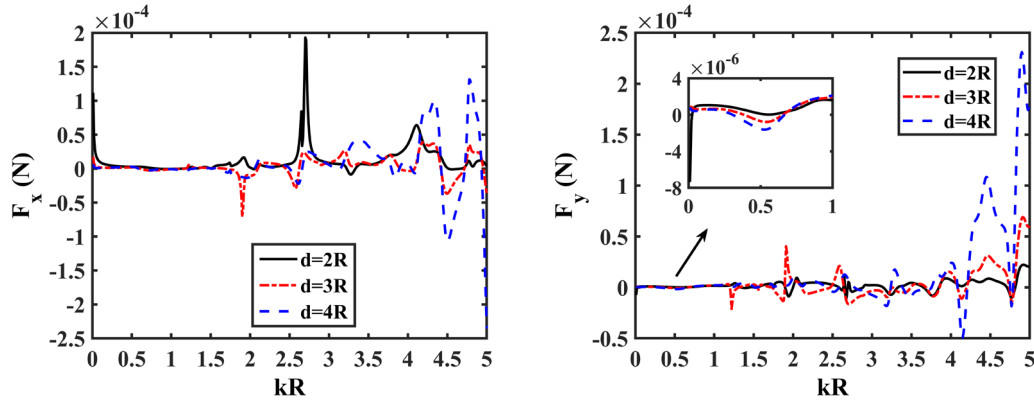


Figure 3. ARFs for a free PE cylinder in the high viscosity liquid (glycerin) at different d for $\theta = \pi/4$. (a) The axial ARF F_x , (b) The transverse ARF F_y .

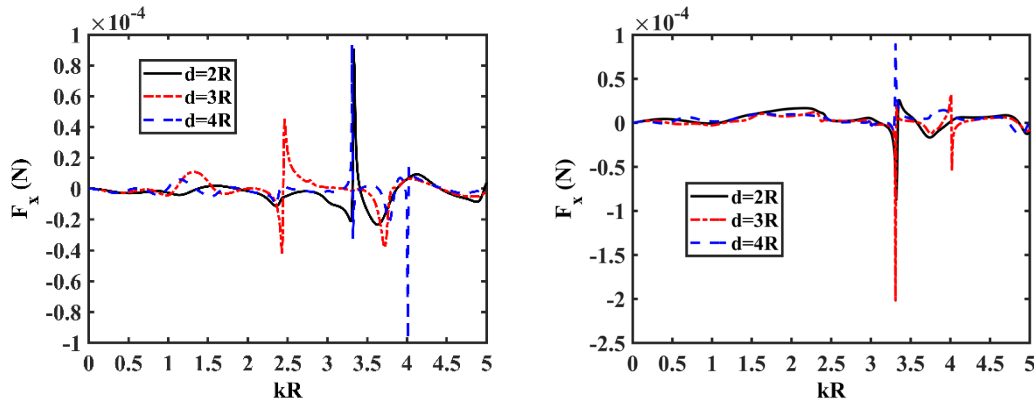


Figure 4. ARFs for a free PE cylinder in the low viscosity liquid (water) at different d for $\theta = \pi/4$. (a) The axial ARF F_x , (b) The transverse ARF F_y .

4. CONCLUSIONS

Manipulation techniques utilizing acoustic radiation force (ARF) hold great promise for applications in diverse fields, including biophysics, ultrasonic medicine etc. So, analysis of the mechanism of the ARF is very important. This paper presents the analytic studies and numerical results of the ARF on a free elastic cylinder in viscous fluids with boundaries under oblique plane wave incidence. The elastic particle suspended freely in a bounded viscous fluid is considered to make it closer to the actual application environment. The scattering coefficients of a free elastic cylinder particle in that sound field defined in the paper are calculated theoretically. The numerical results intuitively show that the mechanism of the ARF is related to the fluid viscosity, plane wave's incident angle, and particle position. The ARFs on an elastic

cylinder versus dimensionless frequency kR exhibit a series of maxima and minima because of its resonant vibrations. The pronounced maxima and minima are mainly emerge on the larger kR . In addition, compared with the ARFs for the particle in the high viscosity liquid (glycerin), the ARFs versus kR for the particle in the low viscosity liquid (water) appear a series of sharp peaks. This study provides deeper insight into the ARF manipulation mechanism and contributes to the enhanced design of control devices.

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APPENDIX

The detailed expressions of the equations in (19) for an elastic cylinder as followings

$$\begin{bmatrix} C_r & D_r \\ C_\varphi & D_\varphi \end{bmatrix} - \begin{bmatrix} S_r & Q_r \\ S_\varphi & Q_\varphi \end{bmatrix} \begin{bmatrix} W_{1n} & O \\ O & W_{2n} \end{bmatrix} \begin{bmatrix} A_n \\ B_n \end{bmatrix} = \begin{bmatrix} I_r \\ I_\varphi \end{bmatrix}. \quad (19)$$

where

$$\begin{aligned} A_n &= \begin{bmatrix} A_{-n}^{(1)} & A_{-n+1}^{(1)} & \cdots & A_{-1}^{(1)} & A_0^{(1)} & A_1^{(1)} & \cdots & A_{n-1}^{(1)} & A_n^{(1)} \end{bmatrix}^T, \\ B_n &= \begin{bmatrix} B_{-n}^{(1)} & B_{-n+1}^{(1)} & \cdots & B_{-1}^{(1)} & B_0^{(1)} & B_1^{(1)} & \cdots & B_{n-1}^{(1)} & B_n^{(1)} \end{bmatrix}^T, \\ W_{1n} &= \begin{bmatrix} (-1)^{-n} H_0^{(1)}(2\alpha d) & (-1)^{-n+1} H_{-1}^{(1)}(2\alpha d) & \cdots & (-1)^n H_{-2n}^{(1)}(2\alpha d) \\ (-1)^{-n} H_1^{(1)}(2\alpha d) & (-1)^{-n+1} H_0^{(1)}(2\alpha d) & \cdots & (-1)^n H_{-2n+1}^{(1)}(2\alpha d) \\ \vdots & \vdots & \ddots & \vdots \\ (-1)^{-n} H_{2n}^{(1)}(2\alpha d) & (-1)^{-n+1} H_{2n-1}^{(1)}(2\alpha d) & \cdots & (-1)^n H_0^{(1)}(2\alpha d) \end{bmatrix}, \\ W_{2n} &= \begin{bmatrix} (-1)^{-n} H_0^{(1)}(2\beta d) & (-1)^{-n+1} H_{-1}^{(1)}(2\beta d) & \cdots & (-1)^n H_{-2n}^{(1)}(2\beta d) \\ (-1)^{-n} H_1^{(1)}(2\beta d) & (-1)^{-n+1} H_0^{(1)}(2\alpha d) & \cdots & (-1)^n H_{-2n+1}^{(1)}(2\beta d) \\ \vdots & \vdots & \ddots & \vdots \\ (-1)^{-n} H_{2n}^{(1)}(2\beta d) & (-1)^{-n+1} H_{2n-1}^{(1)}(2\beta d) & \cdots & (-1)^n H_0^{(1)}(2\beta d) \end{bmatrix}, \\ I_r &= \begin{bmatrix} I_{r,-n} & I_{r,-n+1} & \cdots & I_{r,n} \end{bmatrix}^T, \\ I_\varphi &= \begin{bmatrix} I_{\varphi,-n} & I_{\varphi,-n+1} & \cdots & I_{\varphi,n} \end{bmatrix}^T, \end{aligned}$$

in which

$$\begin{aligned} I_{r,n} &= -2A \cos\left(\alpha d \cos \theta + n\theta - \frac{n\pi}{2}\right) \left\{ \varepsilon_{r,n} (\bar{\beta}_{\varphi,n} \bar{\chi}_{r,n} - \bar{\beta}_{r,n} \bar{\chi}_{\varphi,n}) \right. \\ &\quad \left. + \frac{\mu}{\mu'} [\bar{a}_{r,n} (\beta_{r,n} \bar{\chi}_{\varphi,n} - \beta_{\varphi,n} \bar{\chi}_{r,n}) + \bar{b}_{r,n} (\beta_{\varphi,n} \bar{\beta}_{r,n} - \beta_{r,n} \bar{\beta}_{\varphi,n})] \right\}, \\ I_{\varphi,n} &= -2iA \cos\left(\alpha d \cos \theta + n\theta - \frac{n\pi}{2}\right) \left\{ \varepsilon_{\varphi,n} (\bar{\beta}_{\varphi,n} \bar{\chi}_{r,n} - \bar{\beta}_{r,n} \bar{\chi}_{\varphi,n}) \right. \\ &\quad \left. + \frac{\mu}{\mu'} [\bar{a}_{\varphi,n} (\beta_{r,n} \bar{\chi}_{\varphi,n} - \beta_{\varphi,n} \bar{\chi}_{r,n}) + \bar{b}_{\varphi,n} (\beta_{r,n} \bar{\beta}_{\varphi,n} - \beta_{\varphi,n} \bar{\beta}_{r,n})] \right\}. \end{aligned}$$

In (19), submatrices C_r , C_φ , D_r , D_φ , S_r , S_φ , Q_r and Q_φ are diagonal, with their respective diagonal elements given by $C_{r,n}$, $C_{\varphi,n}$, $D_{r,n}$, $D_{\varphi,n}$, $S_{r,n}$, $S_{\varphi,n}$, $Q_{r,n}$ and $Q_{\varphi,n}$. The matrix O is a zero matrix having the same dimensions as the matrices W_{1n} and W_{2n} . The elements of these diagonal matrices are defined as follows:

$$\begin{aligned}
C_{r,n} &= a_{r,n} (\bar{\beta}_{\varphi,n} \bar{\chi}_{r,n} - \bar{\beta}_{r,n} \bar{\chi}_{\varphi,n}) + \frac{\mu}{\mu'} [\bar{a}_{r,n} (\gamma_{r,n} \bar{\chi}_{\varphi,n} - \gamma_{\varphi,n} \bar{\chi}_{r,n}) + \bar{b}_{r,n} (\gamma_{\varphi,n} \bar{\beta}_{r,n} - \gamma_{r,n} \bar{\beta}_{\varphi,n})], \\
C_{\varphi,n} &= ia_{\varphi,n} (\beta_{\varphi,n} \bar{\chi}_{r,n} - \beta_{r,n} \bar{\chi}_{\varphi,n}) + \frac{i\mu}{\mu'} [\bar{a}_{\varphi,n} (\gamma_{r,n} \bar{\chi}_{\varphi,n} - \gamma_{\varphi,n} \bar{\chi}_{r,n}) + \bar{b}_{\varphi,n} (\gamma_{r,n} \bar{\beta}_{\varphi,n} - \gamma_{\varphi,n} \bar{\beta}_{r,n})], \\
D_{r,n} &= ib_{r,n} (\bar{\beta}_{\varphi,n} \bar{\chi}_{r,n} - \bar{\beta}_{r,n} \bar{\chi}_{\varphi,n}) + \frac{i\mu}{\mu'} [\bar{a}_{r,n} (\delta_{r,n} \bar{\chi}_{\varphi,n} - \delta_{\varphi,n} \bar{\chi}_{r,n}) + \bar{b}_{r,n} (\delta_{\varphi,n} \bar{\beta}_{r,n} - \delta_{r,n} \bar{\beta}_{\varphi,n})], \\
D_{\varphi,n} &= b_{\varphi,n} (\beta_{\varphi,n} \bar{\chi}_{r,n} - \beta_{r,n} \bar{\chi}_{\varphi,n}) + \frac{\mu}{\mu'} [\bar{a}_{\varphi,n} (\delta_{\varphi,n} \bar{\chi}_{r,n} - \delta_{\varphi,n} \bar{\chi}_{\varphi,n}) + \bar{b}_{\varphi,n} (\delta_{\varphi,n} \bar{\beta}_{r,n} - \delta_{r,n} \bar{\beta}_{\varphi,n})], \\
S_{r,n} &= d_{r,n} (\bar{\beta}_{r,n} \bar{\chi}_{\varphi,n} - \bar{\beta}_{\varphi,n} \bar{\chi}_{r,n}) + \frac{\mu}{\mu'} [\bar{a}_{r,n} (\beta_{\varphi,n} \bar{\chi}_{r,n} - \beta_{r,n} \bar{\chi}_{\varphi,n}) + \bar{b}_{r,n} (\beta_{r,n} \bar{\beta}_{\varphi,n} - \beta_{\varphi,n} \bar{\beta}_{r,n})], \\
S_{\varphi,n} &= id_{\varphi,n} (\bar{\beta}_{r,n} \bar{\chi}_{\varphi,n} - \bar{\beta}_{\varphi,n} \bar{\chi}_{r,n}) + \frac{i\mu}{\mu'} [\bar{a}_{\varphi,n} (\beta_{\varphi,n} \bar{\chi}_{r,n} - \beta_{r,n} \bar{\chi}_{\varphi,n}) + \bar{b}_{\varphi,n} (-\beta_{r,n} \bar{\beta}_{\varphi,n} - \beta_{\varphi,n} \bar{\beta}_{r,n})], \\
Q_{r,n} &= ie_{r,n} (\bar{\beta}_{r,n} \bar{\chi}_{\varphi,n} - \bar{\beta}_{\varphi,n} \bar{\chi}_{r,n}) + \frac{i\mu}{\mu'} [\bar{a}_{r,n} (\chi_{\varphi,n} \bar{\chi}_{r,n} - \chi_{r,n} \bar{\chi}_{\varphi,n}) + \bar{b}_{r,n} (\chi_{r,n} \bar{\beta}_{\varphi,n} + \chi_{\varphi,n} \bar{\beta}_{r,n})], \\
Q_{\varphi,n} &= e_{\varphi,n} (\bar{\beta}_{r,n} \bar{\chi}_{\varphi,n} - \bar{\beta}_{\varphi,n} \bar{\chi}_{r,n}) + \frac{\mu}{\mu'} [\bar{a}_{\varphi,n} (\chi_{r,n} \bar{\chi}_{\varphi,n} - \chi_{\varphi,n} \bar{\chi}_{r,n}) + \bar{b}_{\varphi,n} (\chi_{r,n} \bar{\beta}_{\varphi,n} + \chi_{\varphi,n} \bar{\beta}_{r,n})],
\end{aligned}$$

in which

$$\begin{aligned}
a_{r,n} &= (n^2 - n - 0.5\beta^2 R^2) H_n^{(1)}(\alpha R) + \alpha R H_{n+1}^{(1)}(\alpha R), \\
b_{r,n} &= (n^2 - n) H_n^{(1)}(\beta R) - n\beta R H_{n+1}^{(1)}(\beta R), \\
\varepsilon_{r,n} &= (n^2 - n - 0.5\beta^2 R^2) J_n(\alpha R) + \alpha R J_{n+1}(\alpha R), \\
d_{r,n} &= \varepsilon_{r,n}, \\
e_{r,n} &= (n^2 - n) J_n(\beta R) - n\beta R J_{n+1}(\beta R), \\
a_{\varphi,n} &= (n^2 - n) H_n^{(1)}(\alpha R) - n\alpha R H_{n+1}^{(1)}(\alpha R), \\
b_{\varphi,n} &= (n^2 - n + 0.5\beta^2 R^2) H_n^{(1)}(\beta R) - \beta R H_{n+1}^{(1)}(\beta R), \\
\varepsilon_{\varphi,n} &= (n^2 - n) J_n(\alpha R) - n\alpha R J_{n+1}(\alpha R), \\
d_{\varphi,n} &= \varepsilon_{\varphi,n}, \\
e_{\varphi,n} &= (n^2 - n + 0.5\beta^2 R^2) J_n(\beta R) - \beta R J_{n+1}(\beta R), \\
\beta_{r,n} &= n J_n(\alpha R) - \alpha R J_{n+1}(\alpha R),
\end{aligned}$$

$$\gamma_{r,n} = nH_n^{(1)}(\alpha R) - \alpha R_1 H_{n+1}^{(1)}(\alpha R),$$

$$\delta_{r,n} = nH_n^{(1)}(\beta R),$$

$$\chi_{r,n} = nJ_n(\beta R),$$

$$\gamma_{\varphi,n} = nH_n^{(1)}(\alpha R),$$

$$\delta_{\varphi,n} = nH_n^{(1)}(\beta R) - \alpha r_1 H_{n+1}^{(1)}(\beta R),$$

$$\chi_{\varphi,n} = nJ_n(\beta R) - \alpha r_1 J_{n+1}(\beta R)$$

$$\beta_{\varphi,n} = nJ_n(\alpha R),$$

$$\bar{a}_{r,n} = \left(n^2 - n - \bar{k}_l^2 R^2 - \frac{\lambda}{2\mu} \bar{k}_l^2 R^2 \right) J_n(\bar{k}_l R) + \bar{k}_l R J_{n+1}(\bar{k}_l R), \quad \bar{b}_{r,n} = n^2 J_n(\bar{k}_l R) - n \bar{k}_l R J_{n+1}(\bar{k}_l R),$$

$$\bar{a}_{\varphi,n} = (n^2 - n) J_n(\bar{k}_l R) - n \bar{k}_l R J_{n+1}(\bar{k}_l R),$$

$$\bar{b}_{\varphi,n} = (-n^2 + n + \bar{k}_l^2 R^2) J_n(\bar{k}_l R) - \bar{k}_l R J_{n+1}(\bar{k}_l R),$$

$$\bar{\beta}_{r,n} = nJ_n(\bar{k}_l R) - \bar{k}_l R J_{n+1}(\bar{k}_l R),$$

$$\bar{\chi}_{r,n} = nJ_n(\bar{k}_l R),$$

$$\bar{\beta}_{\varphi,n} = nJ_n(\bar{k}_l R),$$

$$\bar{\chi}_{\varphi,n} = nJ_n(\bar{k}_l R) - \bar{k}_l R J_{n+1}(\bar{k}_l R).$$